CSS Colors - Complete Guide

What Are CSS Colors?

Colors in CSS are used to style text, backgrounds, borders, and other elements. CSS supports multiple color formats, each useful for different situations.

1. Color Names

CSS has **147 predefined color names** you can use directly.

Common Color Names

```
color: red;
color: blue;
color: green;
color: yellow;
color: orange;
color: purple;
color: pink;
color: brown;
color: gray;
color: white;
color: black;
```

Extended Color Names

```
color: darkblue;
color: lightblue;
color: skyblue;
color: cornflowerblue;
color: darkgreen;
color: lightgreen;
color: forestgreen;
color: crimson;
color: olive;
color: teal;
```

```
color: navy;
color: maroon;
```

Advantages:

- Easy to remember and read
- No special symbols needed
- Good for quick prototyping

Disadvantages:

- Limited to 147 colors
- No transparency support
- Less precise control

2. Hexadecimal Colors

Hex colors use six characters representing Red, Green, and Blue values (RGB).

Hex Format

```
color: #RRGGBB;

where:
RR = Red (00-FF)
GG = Green (00-FF)
BB = Blue (00-FF)
```

Examples

```
color: #FF0000;         /* Red */
color: #00FF00;         /* Green */
color: #0000FF;         /* Blue */
color: #FFFF00;         /* Yellow (Red + Green) */
color: #FF00FF;         /* Magenta (Red + Blue) */
color: #00FFFF;         /* Cyan (Green + Blue) */
color: #FFFFFF;         /* White (all on) */
color: #000000;         /* Black (all off) */
color: #808080;         /* Gray (all equal) */
color: #FF6347;         /* Tomato */
color: #FFA500;         /* Orange */
```

Shorthand Hex

When each pair has identical digits, use 3-character shorthand:

```
#FF0000 → #F00     /* Red */
#00FF00 → #0F0     /* Green */
#0000FF → #00F     /* Blue */
#FFFF00 → #FF0     /* Yellow */
#FFFFFF → #FFF     /* White */
#000000 → #000     /* Black */

/* Cannot shorten */
#FF6347 → Cannot shorten (different digits)
```

Advantages:

- Most common format
- Unlimited colors (16 million+)
- Widely supported
- Compact notation

Disadvantages:

- Hard to read/remember
- No transparency support
- Requires converting from design tools

3. RGB Colors

RGB uses decimal values for Red, Green, and Blue (0-255).

RGB Format

```
color: rgb(red, green, blue);

where each value is 0-255
```

Examples

```
color: rgb(255, 0, 0);          /* Red */
color: rgb(0, 255, 0);          /* Green */
color: rgb(0, 0, 255);          /* Blue */
color: rgb(255, 255, 0);        /* Yellow */
color: rgb(255, 0, 255);        /* Magenta */
color: rgb(0, 255, 255);        /* Cyan */
color: rgb(255, 255, 255);      /* White */
color: rgb(0, 0, 0);            /* Black */
color: rgb(128, 128, 128);      /* Gray */
color: rgb(255, 99, 71);        /* Tomato */
color: rgb(255, 165, 0);        /* Orange */
```

Advantages:

- More readable than hex
- Easy to understand (0-255 scale)
- Many color pickers output RGB

Disadvantages:

- More verbose than hex
- No transparency support
- Takes more space in code

4. RGBA Colors

RGBA adds an **Alpha channel** (transparency) to RGB.

RGBA Format

```
color: rgba(red, green, blue, alpha);

where:
red, green, blue = 0-255
alpha = 0-1 (0 = fully transparent, 1 = fully opaque)
```

Examples

```
/* Fully opaque (normal) */
color: rgba(255, 0, 0, 1);

/* 75% opaque (25% transparent) */
```

```
color: rgba(255, 0, 0, 0.75);

/* 50% opaque (semi-transparent) */
color: rgba(255, 0, 0, 0.5);

/* 25% opaque (mostly transparent) */
color: rgba(255, 0, 0, 0.25);

/* Fully transparent (invisible) */
color: rgba(255, 0, 0, 0);

/* Useful examples */
background-color: rgba(0, 0, 0, 0.5);    /* Black overlay */
color: rgba(255, 255, 255, 0.9);        /* Almost white text */
border-color: rgba(0, 0, 0, 0.2);       /* Light border */
```

Advantages:

- Supports transparency
- Readable format
- Great for overlays and effects

Disadvantages:

- Verbose syntax
- Less supported in very old browsers

5. HSL Colors

HSL stands for **Hue, Saturation, Lightness**. It's more intuitive than RGB.

HSL Format

```
color: hsl(hue, saturation, lightness);

where:
hue = 0-360 (degrees on color wheel)
saturation = 0-100% (purity of color)
lightness = 0-100% (brightness, 50% is normal)
```

Hue Values

```
0°   = Red
60°  = Yellow
120° = Green
180° = Cyan
240° = Blue
300° = Magenta
360° = Red (back to start)
```

Examples

```
color: hsl(0, 100%, 50%);       /* Red */
color: hsl(120, 100%, 50%);     /* Green */
color: hsl(240, 100%, 50%);     /* Blue */
color: hsl(60, 100%, 50%);      /* Yellow */
color: hsl(0, 0%, 0%);          /* Black */
color: hsl(0, 0%, 100%);        /* White */
color: hsl(0, 0%, 50%);         /* Gray */

/* Darker red (lower lightness) */
color: hsl(0, 100%, 30%);

/* Lighter red (higher lightness) */
color: hsl(0, 100%, 70%);

/* Desaturated red (low saturation) */
color: hsl(0, 20%, 50%);

/* Fully saturated red (high saturation) */
color: hsl(0, 100%, 50%);
```

Advantages:

- Intuitive color selection
- Easy to create color variations
- Great for designers
- Simple to create darker/lighter versions

Disadvantages:

- Less common than hex/RGB
- Requires understanding of hue/saturation/lightness

6. HSLA Colors

HSLA adds **Alpha channel** (transparency) to HSL.

HSLA Format

```
color: hsla(hue, saturation, lightness, alpha);

where alpha = 0-1
```

Examples

```
/* Red with 50% transparency */
color: hsla(0, 100%, 50%, 0.5);

/* Green with 75% opacity */
color: hsla(120, 100%, 50%, 0.75);

/* Transparent overlay */
background-color: hsla(0, 0%, 0%, 0.3);

/* Light text with slight transparency */
color: hsla(0, 0%, 100%, 0.9);
```

7. Hex with Alpha

Newer CSS supports 8-digit hex codes with alpha channel.

Format

```
color: #RRGGBBAA;

where AA is alpha (00-FF)
00 = fully transparent
FF = fully opaque
```

Examples

```
color: #FF0000FF;      /* Red, fully opaque */
color: #FF000080;      /* Red, 50% opaque */
color: #FF000000;      /* Red, fully transparent */
color: #00000080;      /* Black, 50% opaque */
```

```
color: #FFFFFF00;      /* White, fully transparent */

/* Shorthand (when digits match) */
#F00F →   #FF0000FF (red, opaque)
#F008 →   #FF000088 (red, ~53% opaque)
```

8. Special Color Values

Transparent

```
color: transparent;

/* Equivalent to */
color: rgba(0, 0, 0, 0);
```

Current Color

```
/* Uses the element's current text color */
border-color: currentColor;
```

Inherit

```
/* Inherit color from parent */
color: inherit;
```

9. Color Properties

Text Color

```
color: red;
color: #FF0000;
color: rgb(255, 0, 0);
color: hsl(0, 100%, 50%);
```

Background Color

```
background-color: red;
background-color: #FF0000;
background-color: rgba(255, 0, 0, 0.5);
```

Border Color

```
border-color: red;
border-top-color: red;
border-right-color: blue;
border-bottom-color: green;
border-left-color: yellow;
```

Shadow Colors

```
/* Text shadow */
text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.5);

/* Box shadow */
box-shadow: 0 4px 6px rgba(0, 0, 0, 0.3);
```

Outline Color

```
outline-color: red;
```

10. Creating Color Palettes

Monochromatic (Same Hue)

```
/* Different lightness levels of same hue */
primary:   hsl(220, 100%, 50%);
light:     hsl(220, 100%, 70%);
lighter:   hsl(220, 100%, 85%);
dark:      hsl(220, 100%, 30%);
darker:    hsl(220, 100%, 15%);
```

Complementary (Opposite Hue)

```
primary:       hsl(220, 100%, 50%);    /* Blue */
complementary: hsl(40, 100%, 50%);     /* Orange (220 + 180) */
```

Analogous (Adjacent Hues)

```
primary:   hsl(220, 100%, 50%);   /* Blue */
adjacent1: hsl(200, 100%, 50%);   /* Cyan */
adjacent2: hsl(240, 100%, 50%);   /* Purple */
```

11. Color Accessibility

Contrast Ratio

Use sufficient contrast between text and background:

```
/* Bad - low contrast */
color: #999999;
background-color: #CCCCCC;

/* Good - high contrast */
color: #000000;
background-color: #FFFFFF;
```

Don't Rely Only on Color

```
/* Bad - color blind users can't distinguish */
.error { color: red; }
.success { color: green; }

/* Good - add icons or text */
.error { color: red; }
.error::before { content: " "; }
```

WCAG Guidelines

- Level AA: Contrast ratio 4.5:1 for normal text
- Level AAA: Contrast ratio 7:1 for normal text

12. Color Tools

Online Resources

- **Color Picker:** HTML5 input[type="color"]
- **Coolors.co:** Color palette generator
- **Adobe Color Wheel:** Professional color selector
- **WebAIM Contrast Checker:** Accessibility testing

Browser DevTools

- Right-click element → Inspect
- Click color swatch in Styles panel
- Change format (Hex, RGB, HSL)
- Live preview colors

13. Format Comparison

Format	Transparency	Readable	Length
Color name	No	Yes	Short
Hex (#FF0000)	No	No	Short
RGB (rgb())	No	Medium	Medium
RGBA (rgba())	Yes	Medium	Medium
HSL (hsl())	No	Medium	Medium
HSLA (hsla())	Yes	Medium	Medium
Hex + Alpha (#FF0000FF)	Yes	No	Medium

14. Best Practices

1. Use **HSL** for creating color variations
2. Use **RGBA** or **HSLA** for transparency
3. Use **Hex** or **RGB** for consistency
4. Define colors in **CSS variables** for reusability
5. Test colors for **accessibility**
6. Avoid pure black (#000000) or white (#FFFFFF) - use near-black/white
7. Use a **consistent color palette**
8. Document color meanings in comments

CSS Variables for Colors

```
:root {
  --primary-color: #3498db;
  --secondary-color: #2ecc71;
  --error-color: #e74c3c;
  --warning-color: #f39c12;
}
```

```css
p {
  color: var(--primary-color);
}

.button-success {
  background-color: var(--secondary-color);
}

.error-text {
  color: var(--error-color);
}
```

Quick Reference

```css
/* All equivalent to red */
color: red;
color: #FF0000;
color: #F00;
color: rgb(255, 0, 0);
color: rgba(255, 0, 0, 1);
color: hsl(0, 100%, 50%);
color: hsla(0, 100%, 50%, 1);
color: #FF0000FF;
```